Exploring alternative formulations of Smoothed Particle Hydrodynamics for complex fluids and solids

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a Introduction

Clean the mess in the paper, make it look civil

Write proper introduction

Complex fluids and solids can be described by the Symmetric Hyperbolic Thermodynamically Compatible (SHTC) equations. These equations have a Hamiltonian structure and are advantageous in problems with large deformations and shock waves. One of the state variables in the SHTC equations is the distortion field (close to the inverse of the deformation gradient). Recently, a new numerical method solving the SHTC equations was developed based on the Smoothed Particle hydrodynamics (SPH) [2].

However, in a later work [1], an alternative definition of the discrete distortion that is useful when adding noise to the SHTC equations (to be applied in continuum physics of nanomaterials) was developed. In this definition, the discrete distortion is defined as the minimizer of the following functional,

$$D_{\mathbf{x}}(\mathbf{A}) = \sum_a \sum_b w_a(\mathbf{x}) w_b(\mathbf{x}) \|\mathbf{X}_{ab} - \mathbf{A}(\mathbf{x})\mathbf{x}_{ab}\|^2$$

where $\mathbf{X}_{ab}$ is the difference between the Lagrangian positions of the particles a and b, $\mathbf{x}_{ab}$ is the difference between Eulerian positions, and $w_a(\mathbf{x})$ is a weight kernel of particle a (for instance the SPH kernel) and $\mathbf{A}(\mathbf{x})$ is a matrix representation of the distortion field.

This approach is a counterpart to the one used in [2] as we are not evolving the distortion field in time according to the SHTC equations but rather defining it as a minimizer of the functional above at each time step. An advantage of this approach is that it in theory it is possible to add fluctuations to the model, which would enable us to simulate nanomembranes and other nanomaterials, where fluctuations play a crucial role.

b Methods

Add a page explaining notation in use

- distortion as an error estimator (compare different definitions)
- comparison with Ondra's SPH

b.1 Distortion as a minimization problem

In this section we will define the distortion field as a minimizer of the following minimization problem:

$$\mathbf{A}(x) = \underset{\mathbf{A} \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \sum_b w(\mathbf{x} - \mathbf{x}_b) \|\mathbf{A}(\mathbf{x} - \mathbf{x}_b) - (\mathbf{X} - \mathbf{X}_b)\|^2 \tag{1}$$

The term

$$e_b(\mathbf{x}) := \|\mathbf{A}(\mathbf{x} - \mathbf{x}_b) - (\mathbf{X} - \mathbf{X}_b)\| \tag{2}$$

plays the role of the error by which we fail to approximate the linearized relation:

$$\mathbf{A}(x) d\mathbf{x} - d\mathbf{X} = 0 \tag{3}$$

The equation 75 can be than written compactly as:

$$\mathbf{A}(x) = \underset{\mathbf{A} \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \sum_b w(\mathbf{x} - \mathbf{x}_b) e_b(\mathbf{x})^2 \tag{4}$$

which is a weighted least squares problem.

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a = \underset{\mathbf{A} \in \mathbb{R}^{d \times d}}{\operatorname{argmin}} \sum_b w_{ab} \|\mathbf{A}\mathbf{x}_{ab} - \mathbf{X}_{ab}\|^2 \tag{5}$$

Next we will prove that 75 has a solution:

Lemma: (LS problem) 1 *The LS problem 75 has a solution.*

Proof:
Let $I(\mathbf{A}_a) := \frac{1}{2}\sum_b w_{ab}| \ \mathbf{A}_a\mathbf{x}_{ab} - \mathbf{X}_{ab}\|^2$ and observe that $I(\mathbf{A}_a)$ is a differentiable and coercive function meaning:

$$I(\mathbf{A}_a)/\|\mathbf{A}_a\| \to +\infty \tag{6}$$

This implies existence of global minimum.

We can also compute this minimum explicitely:
Lets denote $\mathbf{A}_a\mathbf{x}_{ab} - \mathbf{X}_{ab} =: e_{ab}$ and lets take the Gatteaux derivative of $I(\mathbf{A}_a)$ in the direction $\mathrm{d}\mathbf{A}_a$

$$\begin{aligned}\mathrm{D}I(\mathbf{A}_a)[\mathrm{d}\mathbf{A}_a] := \lim_{\lambda\to 0+}\frac{I(\mathbf{A}_a + \lambda\mathrm{d}\mathbf{A}_a) - I(\mathbf{A}_a)}{\lambda} &= \\ \lim_{\lambda\to 0+}\sum_b w_{ab}\left[(e_{ab} + \lambda\mathrm{d}\mathbf{A}_a\mathbf{x}_{ab}, e_{ab} + \lambda\mathrm{d}\mathbf{A}_a\mathbf{x}_{ab}) - (e_{ab}, e_{ab})\right] &= \\ \lim_{\lambda\to 0+}\frac{1}{2\lambda}\sum_b w_{ab}\left(2(e_{ab}, \lambda\mathrm{d}\mathbf{A}_a\mathbf{x}_{ab}) + \lambda^2\left(\mathrm{d}\mathbf{A}_a\mathbf{x}_a b, \mathrm{d}\mathbf{A}_a\mathbf{x}_a b\right) \right. &= \\ \sum_b w_{ab}\left(\mathbf{A}_a\mathbf{x}_{ab} - \mathbf{X}_{ab}\right)\cdot\mathrm{d}\mathbf{A}_a\mathbf{x}_{ab} &= \\ \mathrm{d}\mathbf{A}_a : \left(\sum_b w_{ab}\left(\mathbf{A}_a\mathbf{x}_{ab} - \mathbf{X}_{ab}\right)\cdot\mathbf{x}_{ab}\right)&\end{aligned} \tag{7}$$

Which immediately implies:

$$\mathbf{A}_a = \mathbf{P}_a\mathbf{Q}_a^{-1} \tag{8}$$

where

$$\mathbf{P}_a = \left(\sum_b w_{ab}\mathbf{X}_{ab}\otimes\mathbf{x}_{ab}\right) \text{ and } \mathbf{Q}_a = \left(\sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right) \tag{9}$$

Notice that the matrix $\mathbf{Q}_a$ is positive definite and hence invertible (unless $\mathbf{x}_{ab} = 0$ for all a, b). To prove it, we need to show that $\mathbf{v}\cdot\mathbf{Q}_a\mathbf{v} > 0$ for each nonzero $\mathbf{v}$.

$$\mathbf{v}\cdot\mathbf{Q}_a\mathbf{v} = \mathbf{v}\cdot\left(\sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)\mathbf{v} = \sum_b w_{ab}(\mathbf{x}_{ab}\cdot\mathbf{v})^2 > 0 \tag{10}$$

We can also generalize this and use the definition from Pavelka & Hutter [1] where we hope to achieve better precision at the cost of summing over all particle pairs instead of singletons.

$$\mathbf{A}^{double}(x) = \underset{\mathbf{A}\in\mathbb{R}^{d\times d}}{\operatorname{argmin}}\sum_b\sum_c w(\mathbf{x}-\mathbf{x}_b)w(\mathbf{x}-\mathbf{x}_c)\|\mathbf{A}\mathbf{x}_{bc} - \mathbf{X}_{bc}\|^2 \tag{11}$$

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a^{double} = \underset{\mathbf{A}\in\mathbb{R}^{d\times d}}{\operatorname{argmin}}\sum_b\sum_c w_{ab}w_{ac}\|\mathbf{A}\mathbf{x}_{bc} - \mathbf{X}_{bc}\|^2 \tag{12}$$

Following the same procedure as in the first example we get the explicit formula for $\mathbf{A}^{double}$ in this case:

$$\mathbf{A}^{double}(\mathbf{x}) = \left(\sum_{a,b} w_a w_b\mathbf{X}_{ab}\otimes\mathbf{x}_{ab}\right)\left(\sum_{a,b} w_a w_b\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)^{-1} \tag{13}$$

Evaluated at position $\mathbf{x}_a$ associated with particle a we get the following:

$$\mathbf{A}_a^{double} = \mathbf{P}_a^{double}(\mathbf{Q}_a^{double})^{-1} \tag{14}$$

where

$$\mathbf{P}_a^{double} = \left(\sum_{b,c} w_{ab}w_{ac}\mathbf{X}_{bc}\otimes\mathbf{x}_{bc}\right) \text{ and } \mathbf{Q}_a^{double} = \left(\sum_{b,c} w_{ab}w_{ac}\mathbf{x}_{bc}\otimes\mathbf{x}_{bc}\right) \tag{15}$$

We can make the following interesting observation that allows us to rewrite $\mathbf{A}^d ouble_a$ in an alternative way:

Notice:

$$\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} = (\mathbf{x}_b - \mathbf{x}_c) \otimes (\mathbf{x}_b - \mathbf{x}_c) = \mathbf{x}_b \otimes \mathbf{x}_b + \mathbf{x}_c \otimes \mathbf{x}_c - \mathbf{x}_b \otimes \mathbf{x}_c - \mathbf{x}_c \otimes \mathbf{x}_b \tag{16}$$

Therefore we have:

$$\begin{aligned} \mathbf{Q}_a^{double} &= \sum_{b,c} w_{ab} w_{ac} \left(\mathbf{x}_b \otimes \mathbf{x}_b + \mathbf{x}_c \otimes \mathbf{x}_c - \mathbf{x}_b \otimes \mathbf{x}_c - \mathbf{x}_c \otimes \mathbf{x}_b \right) = \\ 2 \left(\sum_b w_{ab} \right) &\left(\sum_b w_{ab} \mathbf{x}_b \otimes \mathbf{x}_b \right) - 2 \left(\sum_b w_{ab} \mathbf{x}_b \right) \otimes \left(\sum_b w_{ab} \mathbf{x}_b \right) \end{aligned} \tag{17}$$

which after introducing the notation: $\bar{\mathbf{x}}_a := (\sum_b w_{ab}\mathbf{x}_b)/(\sum_b w_{ab})$ and realizing that $\sum_b w_{ab}(\mathbf{x}_b - \bar{\mathbf{x}}_a) = 0$ can be written as:

$$\mathbf{Q}_a^{double} = 2 \left(\sum_b w_{ab} \right) \left(\sum_b (\mathbf{x}_b - \bar{\mathbf{x}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a) \right) \tag{18}$$

And similarly:

$$\mathbf{P}_a^{double} = 2 \left(\sum_b w_{ab} \right) \left(\sum_b (\mathbf{X}_b - \bar{\mathbf{X}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a) \right) \tag{19}$$

We can also get rid of the prefactor $2\left(\sum_b w_{ab}\right)$ as it will not change the value of $\mathbf{A}_a^{double}$ and simply write:

$$\mathbf{A}_a^{double} = \left(\sum_b (\mathbf{X}_b - \bar{\mathbf{X}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a) \right) \left(\sum_b (\mathbf{x}_b - \bar{\mathbf{x}}_a) \otimes (\mathbf{x}_b - \bar{\mathbf{x}}_a) \right)^{-1} \tag{20}$$

Notice that to compute $\mathbf{A}_a^{double}$ from this definition we only need to know $\bar{\mathbf{x}}_a := (\sum_b w_{ab}\mathbf{x}_b)/(\sum_b w_{ab})$ which can be precomputed.

Next we will make an observation about the order of approximation of $\mathbf{A}_a$ and $\mathbf{A}_a^{double}$: Lets start with the following taylor expansion:

$$\mathbf{X}_b = \mathbf{X}_a + \mathbf{A}(\mathbf{x}_a)\mathbf{x}_{ba} + \frac{1}{2}\nabla \mathbf{A}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] + O(h^3) \tag{21}$$

we can plug it in the definition of $\mathbf{P}_a$ to get:

$$\begin{aligned} \mathbf{P}_a = \sum_b w_{ab} \left(\mathbf{A}(\mathbf{x}_a)\mathbf{x}_{ba} + \frac{1}{2}\nabla \mathbf{A}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] + O(h^3) \right) \otimes \mathbf{x}_{ab} = \\ \mathbf{A}(\mathbf{x}_a) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} + O(h^3) \right) = \mathbf{A}(\mathbf{x}_a)\mathbf{Q}_a + O(h^3) \end{aligned} \tag{22}$$

And finally from the definition of $\mathbf{A}_a$ and using the fact that $\mathbf{Q}_a^{-1} = O(h^2)$ we get:

$$\mathbf{A}_a = \mathbf{P}_a \mathbf{Q}_a^{-1} = \left(\mathbf{A}(\mathbf{x}_a) + O(h^3) \right) \mathbf{Q}_a^{-1} = \mathbf{A}(\mathbf{x}_a) + O(h) \tag{23}$$

For $\mathbf{A}^{double}$ we can proceed similarly (using the notation $\nabla \mathbf{A}^{double}(\mathbf{x}_a) =: \mathbf{H}(\mathbf{x}_a)$):

$$\begin{aligned} \mathbf{X}_{ba} = \mathbf{A}(\mathbf{x}_a)\mathbf{x}_{ba} + \frac{1}{2}\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] + O(h^3) \\ \mathbf{X}_{ca} = \mathbf{A}(\mathbf{x}_a)\mathbf{x}_{ca} + \frac{1}{2}\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ca}, \mathbf{x}_{ca}] + O(h^3) \end{aligned} \tag{24}$$

taking the difference of these two expansions we get:

$$\mathbf{X}_{bc} = \mathbf{A}(\mathbf{x}_a)\mathbf{x}_{bc} + \frac{1}{2}\left(\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] - \mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ca}, \mathbf{x}_{ca}] \right) + O(h^3) \tag{25}$$

pluggin this in the definition of $\mathbf{P}_a^{double}$ and swapping the indecees in the last step we get:

$$\mathbf{P}_a^{double} = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{A}(\mathbf{x}_a) \mathbf{x}_{bc} \otimes \mathbf{x}_{bc} + w_{ab} w_{ac} \frac{1}{2} \left(\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] - \mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ca}, \mathbf{x}_{ca}] \right) \otimes \mathbf{x}_{bc} \right) + O(h^4) =$$
$$\mathbf{A}(\mathbf{x}_a) \mathbf{Q}_a^{double} + \sum_{b,c} w_{ab} w_{ac} \frac{1}{2} \left(\mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ba}, \mathbf{x}_{ba}] - \mathbf{H}(\mathbf{x}_a)[\mathbf{x}_{ca}, \mathbf{x}_{ca}] \right) \otimes \mathbf{x}_{bc} + O(h^4) = \mathbf{A}(\mathbf{x}_a) \mathbf{Q}_a^{double} + O(h^4) \tag{26}$$

Finally we get:

$$\mathbf{A}^{double} = \mathbf{P}_a^{double} (\mathbf{Q}_a^{double})^{-1} = \mathbf{A}(\mathbf{x}_a) + O(h^2) \tag{27}$$

b.2 Particle distortion field

Next we will compute the differential of the distortion field wrt a small perturbation in particle possition. Utilizing the definition of $\mathbf{A}_a$ the differential of the distortion field with respect to position $\mathbf{x}_a$ is then equal to:

$$\mathrm{d}\mathbf{A}_a = \mathrm{d}\mathbf{P}_a \mathbf{Q}_a^{-1} + \mathbf{P}_a \mathrm{d}\mathbf{Q}_a^{-1} = \mathrm{d}\mathbf{P}_a \mathbf{Q}_a^{-1} - \mathbf{P}_a \mathbf{Q}_a^{-1} \mathrm{d}\mathbf{Q}_a \mathbf{Q}_a^{-1} \tag{28}$$

Which under the assumptions from [1]:

$$\dot{\mathbf{x}}_a \approx \dot{\mathbf{x}}_b + \nabla \mathbf{v} \Big|_{\mathbf{x}_a} \cdot \mathbf{x}_{ab} \text{ and } \mathbf{X}_{ab} \approx \mathbf{A}_a \mathbf{x}_{ab} \tag{29}$$

simplifies to:

$$\mathrm{d}\mathbf{A}_a = -\mathbf{A}_a \nabla \mathbf{v}(\mathbf{x}) \Big|_{\mathbf{x}_a} \tag{30}$$

giving us the evolution equation for the distortion field which is the evolution equation for the distortion field in the SHTC equations.
Computing $\mathrm{d}\mathbf{A}_a$ directly without any approximations yields:

$$\begin{aligned} \mathrm{d}\mathbf{A}_a = -\mathbf{A}_a \mathbf{L}_a + \left(\sum_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \left(\mathbf{X}_{ab} - \mathbf{A}_a \mathbf{x}_{ab} \right) \otimes \mathbf{x}_{ab} \right) \mathbf{Q}_a^{-1} + \\ \left(\sum_b w_{ab} \left(\mathbf{X}_{ab} - \mathbf{A}_a \mathbf{x}_{ab} \right) \otimes \mathrm{d}\mathbf{x}_{ab} \right) \mathbf{Q}_a^{-1} \end{aligned} \tag{31}$$

where:

$$\mathbf{L}_a := \left(\sum_b w_{ab} \mathbf{v}_{ab} \otimes \mathbf{x}_{ab} \right) \left(\sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \right)^{-1} \tag{32}$$

The explicit formulas for $\mathrm{d}\mathbf{P}_a$ and $\mathrm{d}\mathbf{Q}_a$ are:

$$\begin{aligned} \mathrm{d}\mathbf{P}_a &= \sum_b w_{ab} \mathbf{X}_{ab} \otimes \mathrm{d}\mathbf{x}_{ab} + \sum_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab} \\ \mathrm{d}\mathbf{Q}_a &= \sum_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab} + \sum_b w_{ab} \mathbf{x}_{ab} \otimes \mathrm{d}\mathbf{x}_{ab} + \sum_b w_{ab} \mathrm{d}\mathbf{x}_{ab} \otimes \mathbf{x}_{ab} \end{aligned} \tag{33}$$

b.3 Equations of motion for A

In this section we will derive the equations of motion.

Density

To proceed we will need the differential of the SPH density

$$\rho_a = \sum_b w_{ab} m_b \tag{34}$$

A straightforward computation yields:

$$\mathrm{d}\rho_a = \sum_b m_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \tag{35}$$

Energy

Let $U = \sum_b \epsilon_b$ be the total energy of the particle system, where ϵ_b is the specific energy density of particle b. We will assume that the specific energy density depends on the density ρ and the distortion field $\mathbf{A}$ ie. $\epsilon_b = \epsilon_b(\rho_b, \mathbf{A}_b)$.

We will also assume that the inner energy ϵ decomposes additively into bulk and shear part as $\epsilon(\rho, \mathbf{A}) = \epsilon_0(\rho) + \epsilon_s(\mathbf{A})$.

We will use the following quadratic relation for the bulk component:

$$\begin{aligned} \epsilon_0(\rho) &= c_0^2/2 \left(\frac{\rho_0}{\rho} - 1\right)^2 \\ \epsilon_0'(\rho) = c_0^2 \left(\frac{\rho_0}{\rho} - 1\right)\left(-\rho_0/\rho^2\right) &= c_0^2 \rho_0 \left(\frac{1}{\rho^2} - \frac{\rho_0}{\rho^3}\right) \end{aligned} \tag{36}$$

For the shear component we will use the following equation:

$$\begin{aligned} \epsilon_s(\mathbf{A}) &= \frac{c_s^2}{4} \|\mathrm{dev}(\mathbf{A}^T\mathbf{A})\|_F^2 \\ \epsilon_s'(\mathbf{A}) &= c_s^2 \mathbf{A}\,\mathrm{dev}(\mathbf{A}^T\mathbf{A}) \end{aligned} \tag{37}$$

Next we will compute the differential of the energy and split it into the bulk part $\mathrm{d}U_0$ and the shear part $\mathrm{d}U_s$ as:

$$\mathrm{d}U = \sum_a m_a \left(\frac{\partial \epsilon_a}{\partial \rho}\mathrm{d}\rho_a + \frac{\partial \epsilon_a}{\partial \mathbf{A}} : \mathrm{d}\mathbf{A}_a\right) = \mathrm{d}U_0 + \mathrm{d}U_s \tag{38}$$

For the bulk part we get the following:

$$\mathrm{d}U_0 = \sum_a m_a \frac{\partial \epsilon_a}{\partial \rho}\mathrm{d}\rho_a = \sum_{a,b} m_a m_b \partial_\rho \epsilon_a \nabla w_{ab} \cdot (\mathrm{d}\mathbf{x}_a - \mathrm{d}\mathbf{x}_b) = \sum_{a,b} m_a m_b \left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b\right) \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_a \tag{39}$$

And for the shear part we will get:

$$\begin{aligned} \mathrm{d}U_s = \sum_a m_a \frac{\partial \epsilon_a}{\partial \mathbf{A}} : \mathrm{d}\mathbf{A}_a = \sum_a m_a \left(\partial_{\mathbf{A}}\epsilon_a : \left(\mathrm{d}\mathbf{P}_a \mathbf{Q}_a^{-1} - \mathbf{P}_a \mathbf{Q}_a^{-1} \mathrm{d}\mathbf{Q}_a \mathbf{Q}_a^{-1}\right)\right) = \\ \sum_a m_a \left(\partial_{\mathbf{A}}\epsilon_a : \mathrm{d}\mathbf{P}_a \mathbf{Q}_a^{-1} - \partial_{\mathbf{A}}\epsilon_a : \mathbf{A}_a \mathrm{d}\mathbf{Q}_a \mathbf{Q}_a^{-1}\right) = \sum_a m_a \left((A) - (B)\right) \end{aligned} \tag{40}$$

The first part labeled (A) yields:

$$\begin{aligned} \sum_a m_a (A) = \sum_a m_a \left[\partial_{\mathbf{A}}\epsilon_a : \mathrm{d}\mathbf{P}_a \mathbf{Q}_a^{-1}\right] = \sum_a m_a \left[\partial_{\mathbf{A}}\epsilon_a \mathbf{Q}_a^{-1} : \mathrm{d}\mathbf{P}_a\right] = \\ \sum_a m_a \left[\mathbf{A}_a^{-T}\mathbf{S}_a : \left(\sum_b w_{ab}\mathbf{X}_{ab} \otimes \mathrm{d}\mathbf{x}_{ab} + \sum_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right)\right] = \\ \sum_a m_a \left[\left(\sum_b w_{ab}\mathbf{S}_a^T \mathbf{A}_a^{-1}\mathbf{X}_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} + \sum_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab}\mathbf{X}_{ab} \cdot \mathbf{A}_a^{-T}\mathbf{S}_a \mathbf{x}_{ab}\right)\right] = \\ \sum_{a,b} m_a \left(w_{ab}\mathbf{S}_a^T \mathbf{A}_a^{-1}\mathbf{X}_{ab} + \nabla w_{ab}\mathbf{A}_a^{-T}\mathbf{S}_a \mathbf{x}_{ab}\right) \cdot (\mathrm{d}\mathbf{x}_a - \mathrm{d}\mathbf{x}_b) = \\ \sum_{a,b} \left[w_{ab}\left(m_a \mathbf{S}_a^T \mathbf{A}_a^{-1} + m_b \mathbf{S}_b^T \mathbf{A}_b^{-1}\right)\mathbf{X}_{ab} + \nabla w_{ab}\mathbf{X}_{ab} \cdot \left(m_a \mathbf{A}_a^{-T}\mathbf{S}_a + m_b \mathbf{A}_b^{-T}\mathbf{S}_b\right)\mathbf{x}_{ab}\right] \cdot \mathrm{d}\mathbf{x}_a \end{aligned} \tag{41}$$

And the second part labeled (B) yields:

$$\begin{aligned}\sum_a m_a(B) = \sum_a m_a \left[\partial_{\mathbf{A}}\epsilon_a : \mathbf{A}_a \mathrm{d}\mathbf{Q}_a\mathbf{Q}_a^{-1}\right] = \sum_a m_a \left[\mathbf{A}_a^T \partial_{\mathbf{A}}\epsilon_a \mathbf{Q}_a^{-1} : \mathrm{d}\mathbf{Q}_a\right] = \sum_a m_a \left[\mathbf{S}_a : \mathrm{d}\mathbf{Q}_a\right] = \\ \sum_a m_a \left[\mathbf{S}_a : \left(\sum_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab}\mathbf{x}_{ab} \otimes \mathbf{x}_{ab} + \sum_b w_{ab}\mathbf{x}_{ab} \otimes \mathrm{d}\mathbf{x}_{ab} + \sum_b w_{ab}\mathrm{d}\mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)\right] = \\ \sum_a m_a \left[\left(\sum_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab}\mathbf{x}_{ab} \cdot \mathbf{S}_a\mathbf{x}_{ab} + \sum_b w_{ab}\mathbf{S}_a^T\mathbf{x}_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} + \sum_b w_{ab}\mathrm{d}\mathbf{x}_{ab} \cdot \mathbf{S}_a\mathbf{x}_{ab}\right)\right] = \\ \sum_a m_a \left[\left(\sum_b \nabla w_{ab}\mathbf{x}_{ab} \cdot \mathbf{S}_a\mathbf{x}_{ab} + \sum_b w_{ab}\mathbf{S}_a^T\mathbf{x}_{ab} + \sum_b w_{ab} \cdot \mathbf{S}_a\mathbf{x}_{ab}\right) \cdot (\mathrm{d}\mathbf{x}_a - \mathrm{d}\mathbf{x}_b)\right] = \\ \sum_{a,b} \left[\nabla w_{ab}\mathbf{x}_{ab} \cdot (m_a\mathbf{S}_a + m_b\mathbf{S}_b)\mathbf{x}_{ab} + w_{ab}\left(m_a\mathbf{S}_a^T + m_b\mathbf{S}_b^T\right)\mathbf{x}_{ab} + w_{ab}(m_a\mathbf{S}_a + m_b\mathbf{S}_b)\mathbf{x}_{ab}\right] \cdot \mathrm{d}\mathbf{x}_a\end{aligned} \tag{42}$$

Finally, putting these terms back together we get:

$$\begin{aligned}\sum_a m_a((A) - (B)) = \\ \sum_{a,b}\left(w_{ab}\left[m_a\mathbf{S}_a^T\left(\mathbf{A}_a^{-1}\mathbf{X}_{ab} - \mathbf{x}_{ab}\right) + m_b\mathbf{S}_b^T\left(\mathbf{A}_b^{-1}\mathbf{X}_{ab} - \mathbf{x}_{ab}\right)\right] + \right. \\ \sum_{a,b}\nabla w_{ab}\left[\left(\mathbf{A}_a^{-1}\mathbf{X}_{ab} - \mathbf{x}_{ab}\right) \cdot (m_a\mathbf{S}_a\mathbf{x}_{ab}) + \left(\mathbf{A}_b^{-1}\mathbf{X}_{ab} - \mathbf{x}_{ab}\right) \cdot (m_b\mathbf{S}_b\mathbf{x}_{ab})\right] + \\ \left. -\sum_{a,b} w_{ab}(m_a\mathbf{S}_a + m_b\mathbf{S}_b)\mathbf{x}_{ab}\right) \cdot \mathrm{d}\mathbf{x}_a\end{aligned} \tag{43}$$

We will further employ the notation $e_{ab} = \mathbf{A}_a^{-1}\mathbf{X}_{ab} - \mathbf{x}_{ab}$ and with that the full differential of energy $\mathrm{d}U$ reads:

$$\begin{aligned}\mathrm{d}U = \sum_{a,b}\left(m_a m_b(\partial_\rho\epsilon_a + \partial_\rho\epsilon_b)\nabla w_{ab} - \sum_{a,b} w_{ab}(m_a\mathbf{S}_a + m_b\mathbf{S}_b)\mathbf{x}_{ab} + \right. \\ \left. w_{ab}\left[m_a\mathbf{S}_a^T e_{ab} - m_b\mathbf{S}_b^T e_{ba}\right] + \sum_{a,b}\nabla w_{ab}\left[e_{ab} \cdot (m_a\mathbf{S}_a\mathbf{x}_{ab}) + e_{ba} \cdot (m_b\mathbf{S}_b\mathbf{x}_{ab})\right]\right) \cdot \mathrm{d}\mathbf{x}_a\end{aligned} \tag{44}$$

Using Newton's second law $m\dot{\mathbf{v}} = -\frac{\mathrm{d}U}{\mathrm{d}\mathbf{x}}$ we get the following equations of motion:

$$\begin{aligned}m_a\dot{\mathbf{v}}_a = -\sum_{a,b}\left(m_a m_b(\partial_\rho\epsilon_a + \partial_\rho\epsilon_b)\nabla w_{ab} - \sum_{a,b} w_{ab}(m_a\mathbf{S}_a + m_b\mathbf{S}_b)\mathbf{x}_{ab} + \right. \\ \left. w_{ab}\left[m_a\mathbf{S}_a^T e_{ab} - m_b\mathbf{S}_b^T e_{ba}\right] + \sum_{a,b}\nabla w_{ab}\left[e_{ab} \cdot (m_a\mathbf{S}_a\mathbf{x}_{ab}) + e_{ba} \cdot (m_b\mathbf{S}_b\mathbf{x}_{ab})\right]\right)\end{aligned} \tag{45}$$

TODO: Add dimensional analysis, the units are the same so we have at least some hope that we ended up with something meaningful

Here we are presented with two options, we can either work with the full system, or we can hope that the contribution of $e_{ab} = \mathbf{A}_a^{-1}\mathbf{X}_{ab} - \mathbf{x}_{ab}a$ will be very small (which should be the case) and we can work with significantly simplier system given by:

$$\dot{\mathbf{x}}_a = \mathbf{v}_a \tag{46a}$$

$$\dot{\mathbf{v}}_a = \sum_b m_b\left(w_{ab}\left(\frac{\mathbf{S}_a}{m_b} + \frac{\mathbf{S}_b}{m_a}\right) - (\partial_\rho\epsilon_a + \partial_\rho\epsilon_b)\nabla w_{ab}\right) \tag{46b}$$

$$\mathbf{A}_a = \left(\sum_b w_{ab}\mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right)\left(\sum_b w_{ab}\mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1} \tag{46c}$$

where

$$\mathbf{S}_a = c_s^2 \mathbf{A}_a^T \mathbf{A}_a \mathrm{dev}(\mathbf{A}_a^T \mathbf{A}_a) \mathbf{Q}_a^{-1} \quad \text{and} \quad \partial_\rho \epsilon_a = c_0^2 \rho_0 \left(\frac{1}{\rho_a^2} + \frac{\rho_0}{\rho_a^3} \right) \tag{46d}$$

Add a statement and proof of the conservational properties

b.3.1 Evolution equations for $\mathbf{A}^{double}$

Here we will try to derive the full evolution equations for the general form of $\mathbf{A}$ from Pavelka a Hutter [1]:

$$\mathbf{A}_a^{double} := \mathbf{P}^{double} \left(\mathbf{Q}^{double} \right)^{-1} = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{X}_{bc} \otimes \mathbf{x}_{bc} \right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right)^{-1} \tag{47}$$

Next, analogously to the previous section we can write the differential of $\mathbf{A}_a$ in the following manner:

$$\mathrm{d}\mathbf{A}_a = \mathrm{d} \left(\mathbf{P}_a^{double} \left(\mathbf{Q}^{double} \right)_a^{-1} \right) = \mathrm{d}\mathbf{P}^{double} \mathbf{Q}^{double} - \mathbf{A}^{double} \mathrm{d}\mathbf{Q}^{double} \left(\mathbf{Q}^{double} \right)^{-1} \tag{48}$$

where $\mathbf{P}_a^{double}$ and $\mathbf{Q}_a^{double}$ are:

$$\begin{aligned} \mathrm{d}\mathbf{P}_a^{double} &= 2 \sum_{b,c} \left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \right) w_{ab} \mathbf{X}_{bc} \otimes \mathbf{x}_{bc} + \sum_{b,c} w_{ab} w_{ac} \mathbf{X}_{bc} \otimes \mathrm{d}\mathbf{x}_{bc} \\ \mathrm{d}\mathbf{Q}_a^{double} &= 2 \sum_{b,c} \left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \right) w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc} + \sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathrm{d}\mathbf{x}_{bc} + \sum_{b,c} w_{ab} w_{ac} \mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \end{aligned} \tag{49}$$

plugging this in the previous expression we get:

$$\begin{aligned} \mathrm{d}\mathbf{A}_a^{double} &= 2 \sum_{b,c} \left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \right) w_{ab} \mathbf{X}_{bc} \otimes \mathbf{x}_{bc} + \sum_{b,c} w_{ab} w_{ac} \mathbf{X}_{bc} \otimes \mathrm{d}\mathbf{x}_{bc} \left(\mathbf{Q}^{double} \right)^{-1} - \\ \mathbf{A}_a^{double} & \left(2 \sum_{b,c} \left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \right) w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc} + \sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathrm{d}\mathbf{x}_{bc} + \sum_{b,c} w_{ab} w_{ac} \mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right) = \\ & 2 \left(\sum_{b,c} \left(\nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \right) w_{ac} \left(\mathbf{X}_{bc} - \mathbf{A}_a^{double} \mathbf{x}_{bc} \right) \otimes \mathbf{x}_{bc} \right) \left(\mathbf{Q}^{double} \right)^{-1} + \\ & \left(\sum_{b,c} w_{ab} w_{ac} \left(\mathbf{X}_{bc} - \mathbf{A}_a^{double} \mathbf{x}_{bc} \right) \otimes \mathrm{d}\mathbf{x}_{bc} \right) \left(\mathbf{Q}^{double} \right)^{-1} + \\ & -\mathbf{A}_a^{double} \left(\sum_{b,c} w_{ab} w_{ac} \mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_b c \right) \left(\mathbf{Q}^{double} \right)^{-1} \end{aligned} \tag{50}$$

We can identify the error term $e_{abc} := \mathbf{X}_{bc} - \mathbf{A}_a^{double} \mathbf{x}_{bc}$ and work with the simplified system in case its small. The simplified differential of $\mathbf{A}^{double}$ is than:

$$\mathrm{d}\mathbf{A}_a^{double} = -\mathbf{A}_a^{double} \left(\sum_{b,c} w_{ab} w_{ac} \mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right)^{-1} \tag{51}$$

Next we will compute the differential of the energy as in the previous section. We will continue as in the case for $\mathbf{A}$ and split it into the bulk part $\mathrm{d}U_0$ and the shear part $\mathrm{d}U_s$ as:

$$\mathrm{d}U = \sum_a m_a \left(\frac{\partial \epsilon_a}{\partial \rho} \mathrm{d}\rho_a + \frac{\partial \epsilon_a}{\partial \mathbf{A}} : \mathrm{d}\mathbf{A}_a^{double} \right) = \mathrm{d}U_0 + \mathrm{d}U_s \tag{52}$$

For the bulk part we get the same expression as for $\mathbf{A}$:

$$\mathrm{d}U_0 = \sum_{a,b} m_a m_b \left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b \right) \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_a \tag{53}$$

And for the shear part we will get:

$$
\begin{aligned}
\mathrm{d}U_s = \sum_a m_a \frac{\partial \epsilon_a}{\partial \mathbf{A}} : \mathrm{d}\mathbf{A}_a^{double} = \sum_a m_a \left(-\partial_{\mathbf{A}} \epsilon_a : \mathbf{A}_a^{double} \mathrm{d}\mathbf{Q}_a^{double} \left(\mathbf{Q}^{double}\right)^{-1} \right) = \\
-\sum_a m_a \left(\mathbf{A}_a^{double}\right)^T \partial_{\mathbf{A}} \epsilon_a \left(\mathbf{Q}^{double}\right)^{-1} : \left(\sum_{b,c} w_{ab} w_{ac} \mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right) = \\
-\sum_a m_a \mathbf{S}_a^{double} : \left(\sum_{b,c} w_{ab} w_{ac} \mathrm{d}\mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right) = \\
-\sum_{a,b,c} m_a w_{ab} w_{ac} \mathbf{S}_a^{double} \mathbf{x}_{bc} \cdot \mathrm{d}\mathbf{x}_{bc} = -\sum_{a,b,c} m_a w_{ab} w_{ac} \mathbf{S}_a^{double} \mathbf{x}_{bc} \cdot \mathrm{d}\mathbf{x}_b + \sum_{a,b,c} m_a w_{ab} w_{ac} \mathbf{S}_a^{double} \mathbf{x}_{bc} \cdot \mathrm{d}\mathbf{x}_c = \\
-\sum_{b,a,c} m_b w_{ab} w_{bc} \mathbf{S}_b^{double} \mathbf{x}_{ac} \cdot \mathrm{d}\mathbf{x}_a + \sum_{c,b,a} m_c w_{cb} w_{ac} \mathbf{S}_c^{double} \mathbf{x}_{ba} \cdot \mathrm{d}\mathbf{x}_a = \\
\sum_{a,b,c} \left(m_b w_{ab} w_{bc} \mathbf{S}_b^{double} \mathbf{x}_{ca} + m_c w_{cb} w_{ac} \mathbf{S}_c^{double} \mathbf{x}_{ba} \right) \cdot \mathrm{d}\mathbf{x}_a = 2 \left(\sum_{a,b,c} m_b w_{bc} w_{ab} \mathbf{S}_b^{double} \mathbf{x}_{ca} \right) \cdot \mathrm{d}\mathbf{x}_a = \\
2 \left(\sum_{a,b} m_b w_{ab} \mathbf{S}_b^{double} \left(\sum_c w_{bc} \mathbf{x}_c - \sum_c w_{bc} \mathbf{x}_a \right) \right) \cdot \mathrm{d}\mathbf{x}_a = \left(\sum_{a,b} m_b w_{ab} 2 W_a \mathbf{S}_b^{double} \left(\bar{\mathbf{x}}_b - \mathbf{x}_a \right) \right) \cdot \mathrm{d}\mathbf{x}_a
\end{aligned}
\tag{54}
$$

Finally, using Newton's second law $m\dot{\mathbf{v}} = -\frac{\mathrm{d}U}{\mathrm{d}\mathbf{x}}$ we get the following equations of motion for $\mathbf{A}_a^{double}$:

$$
m_a \dot{\mathbf{v}}_a = -\sum_b \left(m_a m_b \left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b \right) \nabla w_{ab} - m_b w_{ab} 2 W_a \mathbf{S}_b^{double} \left(\mathbf{x}_a - \bar{\mathbf{x}}_b \right) \right) \tag{55}
$$

And the full simplified system of equations for $\mathbf{A}^{double}$ reads as:

Add the non simplified force equation

$$
\dot{\mathbf{x}}_a = \mathbf{v}_a \tag{56a}
$$

$$
\dot{\mathbf{v}}_a = -\sum_b m_b \Big(\left(\partial_\rho \epsilon_a + \partial_\rho \epsilon_b \right) \nabla w_{ab} + \frac{2}{m_b} w_{ab} W_a \mathbf{S}_b^{double} \left(\mathbf{x}_a - \bar{\mathbf{x}}_b \right) \Big) \tag{56b}
$$

$$
\mathbf{A}_a^{double} := \mathbf{P}^{double} \left(\mathbf{Q}^{double}\right)^{-1} = \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{X}_{bc} \otimes \mathbf{x}_{bc} \right) \left(\sum_{b,c} w_{ab} w_{ac} \mathbf{x}_{bc} \otimes \mathbf{x}_{bc} \right)^{-1} \tag{56c}
$$

where

$$
\begin{aligned}
\mathbf{S}_a^{double} = c_s^2 \left(\mathbf{A}_a^{double}\right)^T \mathbf{A}_a^{double} \mathrm{dev} \left(\left(\mathbf{A}_a^{double}\right)^T \mathbf{A}_a^{double} \right) \left(\mathbf{Q}^{double}\right)_a^{-1} \\
\partial_\rho \epsilon_a = c_0^2 \rho_0 \left(\frac{1}{\rho_a^2} + \frac{\rho_0}{\rho_a^3} \right)
\end{aligned}
\tag{56d}
$$

It is worth comparing the shear terms for both $\mathbf{A}_a$ and $\mathbf{A}_a^{double}$. The shear force in the first case depends on the pairwise distance $\mathbf{x}_a - \mathbf{x}_b$ and in the second case it depends on the distance of particle a from the center of mass of its neighbours $\mathbf{x}_a - \bar{\mathbf{x}}_b$ which in return yields one higher order convergence rate.

Remark on conservational properties for $\mathbf{A}^{double}$

b.4 Numerics

In this section we test our previous findings. We are presented with two possible options on how to proceed.

Firstly, we can work directly with $\mathbf{A}_a$ from the the minimalization problem:

$$\mathbf{A}_a := \mathbf{P}_a\mathbf{Q}_a^{-1} = \left(\sum_b w_{ab}\mathbf{X}_{ab}\otimes\mathbf{x}_{ab}\right)\left(\sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)^{-1} \tag{57}$$

compute it each time step and get the relation for force, for $\mathbf{A}_a^{double}$, we can proceed equivalently.

Secondly, we can make use of the relation $\mathrm{d}\mathbf{A} = -\mathbf{AL}$ and instead of computing $\mathbf{A}$ directly, we will compute $\mathbf{L}$ each time step and update $\mathbf{A}$ accordingly.

Recall that for $\mathbf{A}_a$ we have:

$$\mathbf{L}_a := \left(\sum_b w_{ab}\mathbf{v}_{ab}\otimes\mathbf{x}_{ab}\right)\left(\sum_b w_{ab}\mathbf{x}_{ab}\otimes\mathbf{x}_{ab}\right)^{-1} \tag{58}$$

And similarly for $\mathbf{A}_a^{double}$ we have:

$$\mathbf{L}_a^{double} = \left(\sum_{b,c} w_{ab}w_{ac}\mathbf{v}_{bc}\otimes\mathbf{x}_{bc}\right)\left(\sum_{b,c} w_{ab}w_{ac}\mathbf{x}_{bc}\otimes\mathbf{x}_{bc}\right)^{-1} \tag{59}$$

We are than presented with the following system of ODEs:

$$\dot{\mathbf{x}}_a = \mathbf{v}_a \tag{60a}$$

$$m_a\dot{\mathbf{v}}_a = -\mathbf{f}_a \tag{60b}$$

$$\dot{\mathbf{A}}_a = -\mathbf{A}_a\mathbf{L}_a \quad \text{or} \quad (\dot{\mathbf{A}_a^{double}}) = -\mathbf{A}_a^{double}\mathbf{L}_a^{double} \tag{60c}$$

To solve it we proceed as sugested by Kincl [2] utilizing the following combination of midpoint-rule (for $\mathbf{A}$) and Verlet integration scheme (for $\mathbf{x}$ and $\mathbf{v}$):

$$\begin{aligned}
\mathbf{v}_a(t_{k+1/2}) &= \mathbf{v}_a(t_k) + \frac{\Delta t}{2m_a}\mathbf{f}_a(t_k)\\
\mathbf{x}_a(t_k) &= \mathbf{x}_a(t_k) + \frac{\Delta t}{2}\mathbf{v}_a(t_{k+1/2})\\
\mathbf{A}_a(t_{k+1}) = \mathbf{A}_a(t_k)\left(\mathbf{I} - \frac{\Delta t}{2}\mathbf{L}_a(t_{k+1/2})\right)&\left(\mathbf{I} + \frac{\Delta t}{2}\mathbf{L}_a(t_{k+1/2})\right)^{-1}\\
\mathbf{x}_a(t_{k+1}) &= \mathbf{x}_a(t_{k+1/2}) + \frac{\Delta t}{2}\mathbf{v}_a(t_{k+1/2})\\
\mathbf{v}_a(t_{k+1}) &= \mathbf{v}_a(t_{k+1/2}) + \frac{\Delta t}{2m_a}\mathbf{f}_a(t_{k+1})
\end{aligned} \tag{61}$$

where $t_k = k\Delta t$ denotes the k-th time step and

$$\mathbf{f}_a(t) = \mathbf{f}_a(\mathbf{x}(t), \mathbf{A}(t)), \quad \mathbf{L}_a = \mathbf{L}_a(\mathbf{x}(t), \mathbf{v}(t)) \tag{62}$$

b.5 Beryllium plate

All the definitions of distortion field $\mathbf{A}$ were tested on Beryllium plate benchmark.

Add the description of Beryllium plate benchmark with parameters

Add pictures showing benchmark failure for the recomputed case

Add pictures showing benchmark working for the evolved case

Add figures with energy

Add figures with amplitude matching the expected results

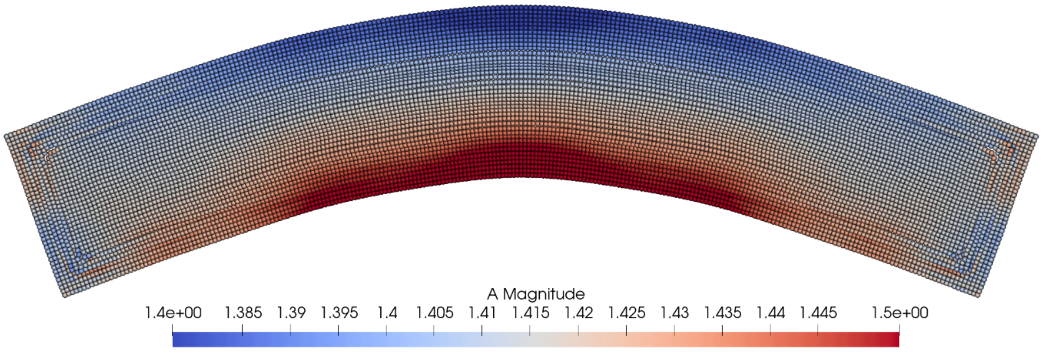

Figure 1: For small deformations and large support radius it seems to work at first.

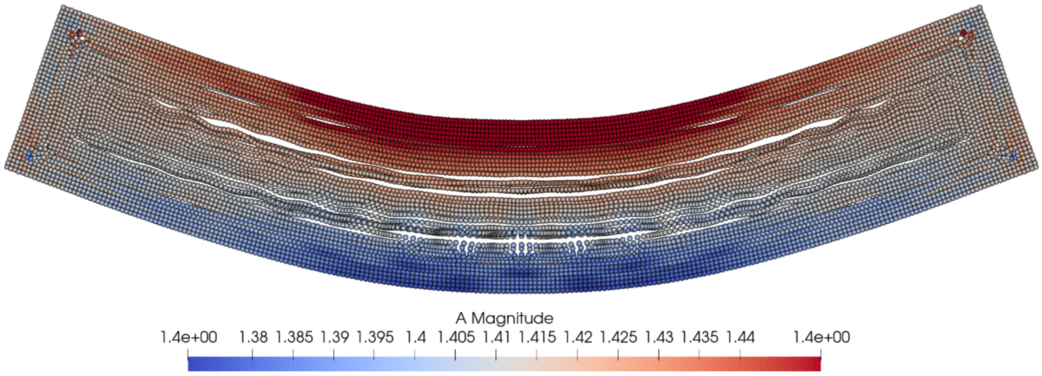

Figure 2: However, even with support radius $h = 20dr$ (which is huge) the simulation becomes unstable after some time. This does not seem to be tensile instability as the pressure is positive in the problematic places. It more has to do with this approximation $\mathbf{X} - \mathbf{A}\mathbf{x} \approx 0$.

b.5.1 Comparison with Kincl's framework

In this section we compare the main conceptual differences between the SHTC-SPH method developed by Kincl et al. MLS methods and between our work. Two disting approaches to approximating $\nabla \mathbf{v}$ arise in meshfree methods. Both approaches are equally valid and internally consistent.

SPH:

Lets first review the approach utilized by SPH. Recall that at the heart of SPH lies the idea of approximating differential operators and scalar functions by first convoluting them with the mollifiying kernel w:

$$\begin{aligned} f = f * \delta \approx f * w \\ \nabla f = \nabla f * \approx \nabla f * w = f * \nabla w \end{aligned} \tag{63}$$

and than approximating the resulting integral with numerical quadrature:

$$\begin{aligned} (f * w)(\mathbf{x}_a) = \int f(\mathbf{y}) w(\mathbf{x}_a - \mathbf{y}) \mathrm{d}\mathbf{y} \approx \sum_b V_b f(\mathbf{x}_b) w_{ab} \\ (f * \nabla w)(\mathbf{x}_a) = \int f(\mathbf{y}) \nabla w(\mathbf{x}_a - \mathbf{y}) \mathrm{d}\mathbf{y} = \sum_b V_b f(\mathbf{x}_b) \nabla w_{ab} \end{aligned} \tag{64}$$

using the volumes $V_b = \rho_b / m_b$ of the particles as quadrature weights. Similarly for a vector function

$\mathbf{f}$ we have:

$$\mathbf{f}(\mathbf{x}_a) \approx \sum_b V_b \mathbf{f}(\mathbf{x}_b) w_{ab}$$
$$\nabla \mathbf{f}(\mathbf{x}_a) \approx \sum_b V_b \mathbf{f}(\mathbf{x}_b) \otimes \nabla w_{ab} \tag{65}$$

Recall, that our aim is to get discrete SPH form of the following relation:

$$\mathrm{d}\mathbf{A} = -\mathbf{A}\mathrm{d}\mathbf{L} \tag{66}$$

Following the SPH philosophy we can approximate $\mathbf{L} = \nabla \mathbf{v}$ directly as:

$$\mathbf{L} = \nabla \mathbf{v} = \sum_b V_b \mathbf{v}_{ab} \otimes \nabla w_{ab} \tag{67}$$

However this discretization is not first order exact. To alleviate this issue

Add the description of the corrected version of SPH velocity gradient

Kincl than uses slightly modified version replacing the volumes V_a with m_a masses of respective particles:

$$\mathbf{L} = \sum_b m_b \mathbf{v}_{ab} \otimes \nabla w_{ab} \tag{68}$$

for a benchmark problem such as Beryllium plate where density is roughly constant $\rho \approx \text{const.}$ this is equivalent to the standard this interplays nicely with the differential of density $\mathrm{d}\rho$:

$$\mathrm{d}\rho_a = \sum_b m_b \nabla w_{ab} \cdot \mathrm{d}\mathbf{x}_{ab} \tag{69}$$

Equations of motion follow from the same procedure as in previous sections and we recover rather elegant system:

$$\dot{\mathbf{x}}_a = \mathbf{v}_a \tag{70a}$$
$$\dot{\mathbf{v}}_a = \sum_b m_b \left(\mathbf{T}_a + \mathbf{T}_b\right) \nabla w_{ab} \tag{70b}$$
$$\dot{\mathbf{A}}_a = -\mathbf{A}_a \mathbf{L}_a \tag{70c}$$

where

$$\mathbf{T}_a := -\partial_\rho \epsilon \mathbf{I} + \mathbf{A}_a^T \partial_{\mathbf{A}} \epsilon \left(\sum_b m_b \mathbf{x}_{ab} \otimes \nabla w_{ab} \right)^{-1} \tag{70d}$$

MLS:

The other approach (refered to as MLS - moving least squares) defines the gradient of a function f as the best affine approximation of it in the sense of a least squares problem

$$\nabla f(\mathbf{x}) = \underset{\mathbf{A} \in \mathbb{R}^{d\times d}}{\operatorname{argmin}} \int w(\mathbf{x}-\mathbf{y}) \| f(\mathbf{y}) - f(\mathbf{x}) - \mathbf{A}\cdot(\mathbf{x}-\mathbf{y})\|^2 \mathrm{d}\mathbf{y} \tag{71}$$

which gives:

$$\nabla f(\mathbf{x}) = \left(\int w(\mathbf{x}-\mathbf{y})(f(\mathbf{y}) - f(\mathbf{x})) \otimes (\mathbf{y}-\mathbf{x}) \mathrm{d}\mathbf{y} \right) \left(\int w(\mathbf{x}-\mathbf{y})(\mathbf{y}-\mathbf{x}) \otimes (\mathbf{y}-\mathbf{x}) \mathrm{d}\mathbf{y} \right)^{-1} \tag{72}$$

Motivated by MLS approach we will define the continuum equivallent of our minimalization problem.

$$\mathbf{A}(x) = \underset{\mathbf{A} \in \mathbb{R}^{d\times d}}{\operatorname{argmin}} \sum_b w(\mathbf{x}-\mathbf{x}_b) \|\mathbf{A}(\mathbf{x}-\mathbf{x}_b) - (\mathbf{X}-\mathbf{X}_b)\|^2 \tag{73}$$

which reads as:

$$\mathbf{A}(x) = \underset{\mathbf{A} \in \mathbb{R}^{d\times d}}{\operatorname{argmin}} \int w(\mathbf{x}-\mathbf{y}) \|\mathbf{A}(\mathbf{x}-\mathbf{y}) - (\mathbf{X}-\mathbf{Y})\|^2 \mathrm{d}\mathbf{y} \tag{74}$$

and the SPH discretization we will get the following:

$$\mathbf{A}(x) = \underset{\mathbf{A} \in \mathbb{R}^{d\times d}}{\operatorname{argmin}} \sum_b V_b w(\mathbf{x}-\mathbf{x}_b) \|\mathbf{A}(\mathbf{x}-\mathbf{x}_b) - (\mathbf{X}-\mathbf{X}_b)\|^2 \tag{75}$$

We already now the solution of this problem:

$$\mathbf{A}_a = \left(\sum_b \frac{m_b}{\rho_b} w_{ab} \mathbf{X}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b \frac{m_b}{\rho_b} w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1} \tag{76}$$

Using our previous calculations we also immediately see that:

$$\mathrm{d}\mathbf{A}_a = -\mathbf{A}_a \mathrm{d}\mathbf{L}_a \quad \text{where} \quad \mathrm{d}\mathbf{L}_a = \left(\sum_b V_b w_{ab} \mathrm{d}\mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right) \left(\sum_b V_b w_{ab} \mathbf{x}_{ab} \otimes \mathbf{x}_{ab}\right)^{-1} \tag{77}$$

Pavelka-Hutter:

Finish this section

Lastly we have the least squares particle estimator defined in [1]. Notice that unlike the two previous this definition does not factor in the varying volumes of different particles and is purely geometrical.

For problems where the density remains roughly constant this is of no issue, however it might be the case that this definition might not be suitable for highly compressible problems with changing volume.

Add a table containing the different frameworks (MLS, SHTC-SPH, Pavelka-Hutter) and write down how the different quantities ($\mathbf{A}$, $\mathbf{L}$, energy) look like

c Conclusions

Write down a proper conclusion

We have successfully implemented and tested the new definition for the distortion field as defined in Pavelka and Hutter. WE have also compared the three different definitions of distortion field arrising from several distinct discretization philosophies. We have tested the distinct formulations on a beryllium plate benchmark and we tested two different integration schemes. The scheme based on recomputing the distortion field every iteration proved to be unstable. The scheme based on the continuum identity $\dot{\mathbf{A}} = -\mathbf{A}\mathbf{L}$ showed remarkable conservational properties.

Add references

Add citations

References

[1] M. Hütter and M. Pavelka. Particle-based approach to the eulerian distortion field and its dynamics. *Continuum Mechanics and Thermodynamics*, 35(5):1943– 1967, 2023.

[2] O. Kincl, I. Peshkov, M. Pavelka, and V. Klika. Unified description of fluids and solids in smoothed particle hydrodynamics. *Applied Mathematics and Computation*, 439:127579, 2023.